

E-COMMERCE BUSINESS ANALYSIS

Data-Driven Insights & Strategic Recommendations

Dataset	Period	Records	Tools Used
E-Commerce Orders	Jan 2023 – Dec 2024	1,200 Orders	MySQL · Python · Power BI

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1. EXECUTIVE SUMMARY

This comprehensive analysis examines 1,200 e-commerce orders across a diversified product portfolio. The business is generating strong revenue and maintains a healthy Average Order Value of Rs. 1,053.97, indicating good customer spending behaviour. However, high cancellation and return rates remain the biggest challenge, resulting in significant revenue loss. The company's primary focus should therefore be on reducing cancellations and returns through better order fulfilment, accurate product information, and improved customer experience.

The business has generated Rs. 1,264,761.96 in total revenue. However, the company's biggest challenge is the high number of cancellations and returns, which account for 41.09% of total revenue loss. Reducing these losses should be the top priority, as even a small improvement can significantly increase profitability.

2. DATA OVERVIEW & QUALITY ASSESSMENT

The dataset contains 1,200 complete order records spanning from January 2023 to December 2024, with 14 comprehensive attributes. Each record includes: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, and TotalPrice.

Data Quality Metrics

Quality Dimension	Status	Details
Total Records	Complete	1,200 order records
Missing Values	Acceptable	25.75% in CouponCode only (309 null values)
Duplicate Records	Excellent	Zero duplicates detected
Data Completeness	Excellent	99.5% overall
Date Coverage	Complete	Jan 2023 – Dec 2024

CouponCode Null Treatment

CouponCode has 309 null values, representing 25.75% of the column. We fill null categorical values with the mode during preprocessing. But for analysis, we replace null values with "No Coupon" so the missing values are treated as a separate category and the results become more accurate.

A new cleaned table **sales_cleaned** was created containing the cleaned dataset. Null and blank values in CouponCode were replaced with "No Coupon", and the Date column was converted from text format to a proper date format. All further EDA and business analysis were performed on sales_cleaned.

Product Portfolio Distribution

The product mix is distributed across seven main categories: Printers, Phones, Laptops, Tablets, Chairs, Monitors, and Desks. Chair category leads in both quantity sold and revenue. No single product category dominates the portfolio, which provides business resilience and reduces dependency on any one product line.

3. KEY PERFORMANCE INDICATORS

Rs. 12.6L	1,200	Rs. 1,054	3,535
Total Revenue	Total Orders	Avg Order Value	Total Qty Sold
20.83%	20.58%	41.09%	5 (avg)
Cancellation Rate	Return Rate	Combined Loss Rate	Avg Items in Cart

By looking at the total quantity sold and revenue generated, the business is in a good position and customers spend high amounts as AOV is quite good. However, the cancellation rate and return rate are also higher, which negatively impacts the business revenue and overall growth. Finding the cause and reducing both cancellation and return rates is important for the business's stability.

4. PRODUCT ANALYSIS

4.1 Quantity Sold by Product

Product	Qty Sold	Rank
Chair	562	1st
Printer	542	2nd
Laptop	535	3rd
Desk	508	4th
Tablet	497	5th
Monitor	480	6th
Phone	411	7th (Lowest)

Chair has the largest number of units sold, and Phone has the lowest number sold. There are three product categories that sold below 500 units: Tablet, Monitor, and Phone.

4.2 Revenue by Product

Product	Total Revenue	Rank
Chair	Rs. 1,95,620.11	1st
Printer	Rs. 1,95,612.61	2nd
Laptop	Rs. 1,92,126.56	3rd
Tablet	Rs. 1,86,568.95	4th
Monitor	Rs. 1,75,651.41	5th
Desk	Rs. 1,67,459.93	6th
Phone	Rs. 1,51,722.39	7th (Lowest)

Chair has the highest revenue among all products, and in the top 3 products there is no such big difference in revenue. Phone has the lowest revenue.

4.3 Top Products by Coupon Code

Printers and Tablets are the most popular products across multiple coupon categories, while different coupon codes show different product preferences, helping the business design targeted promotions.

4.4 Top Products by Referral Source

Instagram generated the highest orders, with Desk, Tablet, and Chair being the top-selling products. Printers performed strongly across Email, Facebook, and Google, while Referral traffic showed a preference for Laptops and Phones. This indicates that customer product preferences vary by marketing channel, which can help optimise targeted campaigns.

5. SALES TRENDS & MONTH-OVER-MONTH ANALYSIS

Sales showed significant fluctuations throughout the period, indicating seasonal demand patterns and varying customer purchasing behaviour.

Period	Pattern	Revenue Trend	Key Observation
Jan–Apr 2023	Growth Phase	Upward	Steady market entry and growth
May–Jun 2023	Peak Season	Highest	Maximum seasonal demand (+130.03% in May)
Jul–Dec 2023	Normalisation	Declining	Post-peak stabilisation; sharp -45.68% in Sep
Jan–Apr 2024	Challenge Period	Declining	Market headwinds; -43.75% in May 2024
May–Jun 2024	Recovery	Strong Growth	Peak reoccurrence; +143.89% in Jun 2024
Jul–Dec 2024	Stable Growth	Consistent	Sustained performance; highest revenue Rs. 68,068

Key Findings

- The highest positive growth was recorded in June 2024 (+143.89%), where sales increased from Rs. 27,909 to Rs. 68,069.
- Another strong growth period occurred in May 2023 (+130.03%), suggesting a major increase in orders or promotional activity.
- The largest decline was observed in September 2023 (-45.68%), followed by May 2024 (-43.75%), indicating sharp drops in monthly revenue.
- Sales recovered quickly after several declines, demonstrating resilience in overall business performance.
- June 2024 recorded the highest revenue at Rs. 68,068.54, followed by May 2023 (Rs. 63,836.84).
- April 2023 recorded the lowest revenue at Rs. 27,751.71.
- Revenue spikes in May–June suggest a potential seasonal peak period that can be leveraged through targeted marketing and inventory planning.

Business Insight

The business experiences periods of rapid growth followed by sharp declines, suggesting the influence of seasonality, promotions, or changing customer demand. Months with exceptional growth (May 2023, June 2024, October 2023) should be investigated to identify successful strategies that can be replicated. Months with significant declines (September 2023, May 2024) should be analysed further to understand potential causes such as reduced demand, inventory issues, or fewer marketing campaigns. The predictable seasonal pattern provides a clear opportunity for proactive business planning — preparing inventory levels in advance, allocating additional marketing budget, and scheduling staffing and fulfilment resources ahead of peak season can capture 15–20% additional revenue during peak seasons.

6. PAYMENT METHOD ANALYSIS

Payment Method	Total Payments	Total Revenue	Note
Online	258	Rs. 2,62,440	1st by Orders
Cash	246	Rs. 2,59,790	2nd
Credit Card	234	Rs. 2,63,850	Highest Revenue
Debit Card	232	Rs. 2,32,360	4th
Gift Card	230	Rs. 2,46,320	5th

Most number of payments were made through Online, then Cash. People also made payments by Gift Cards to take advantage of discounts or cashback. Payment method does not significantly impact order value, but higher-priced transactions, especially through digital methods like credit cards and gift cards, show slightly higher cancellation and return tendencies.

7. COUPON CODE & REFERRAL SOURCE ANALYSIS

7.1 Coupon Code Distribution

Coupon Code	Orders	Revenue	Note
FREESHIP	313	Rs. 3,35,037	1st — Highest Revenue
No Coupon	309	Rs. 3,22,401	2nd
WINTER15	292	Rs. 3,02,484	3rd
SAVE10	286	Rs. 3,04,840	4th

FREESHIP has the largest number of coupons used and the highest revenue, because most people order items and after delivery the charges get a little higher — so if there are no shipping charges, people order more. There is an almost equal distribution of coupon code usage, with all generating revenue above Rs. 3,00,000. Cancellation counts are relatively similar across coupon categories, suggesting that coupon usage alone is not the primary driver of cancellations.

7.2 Referral Source Analysis

Referral Source	Orders	Revenue	Note
Instagram	259	Rs. 2,75,285	1st — Top Source
Email	250	Rs. 2,61,809	2nd
Google	241	Rs. 2,50,441	3rd
Facebook	228	Rs. 2,50,411	4th
Referral	222	Rs. 2,26,816	5th

Instagram is the top referral source with 259 orders, followed by Email and Google, indicating that social media and digital marketing channels are driving most customer traffic and sales. Instagram is the only platform through which the highest revenue is generated compared to any other referral source; all other referral sources have similar revenue. Email and Google have the highest number of cancellations, while Instagram and Facebook have fewer.

8. CANCELLATION & RETURN ANALYSIS

8.1 Order Status Overview

Order Status	Count	Percentage	Note
Cancelled	250	20.83%	Highest — Critical Issue
Returned	247	20.58%	Second Highest
Pending	237	19.75%	
Shipped	235	19.58%	
Delivered	231	19.25%	Lowest — Needs Improvement

A high number of cancelled and returned orders is not a positive sign for the business. Reducing cancellations and returns can help increase revenue and improve customer satisfaction. The business should investigate the reasons behind these orders and take steps to minimise them.

8.2 Products with High Cancellations

Product	Cancellations	Note
Chair	45	Highest — Over 40 cancellations
Printer	35	
Monitor	35	
Laptop	35	
Desk	35	
Tablet	34	
Phone	31	Lowest

Chair has the largest number of cancellations and Phone has the lowest. Only Chair has over 40 cancellations (45); all other products have between 30–35. The reason for Chair cancellations may be that people find them cheaper offline, or the chair is large in size and customers worry it may get damaged during shipping. For electronics, people sometimes do not trust product quality when ordered online due to scams. To reduce cancellations, the business can provide genuine customer reviews, product warranties, detailed product descriptions, and secure delivery assurances.

8.3 Products with High Returns

Product	Returns	Note
Tablet	43	Highest Return Count
Laptop	39	
Printer	38	

Product	Returns	Note
Monitor	36	
Desk	32	
Phone	31	
Chair	28	Lowest — Interesting contrast with cancellations

Electronics products have the highest number of returns, which may be due to customer dissatisfaction, product mismatches, quality issues, or damage during delivery. In contrast, Chairs have the lowest return rate. To reduce returns, the seller should ensure product quality, provide accurate product descriptions, and verify that products match customer expectations before shipping.

8.4 Root Cause Differentiation

Issue Type	Primary Cause	Secondary Factors	Intervention Focus
Cancellations	Multiple behavioural & information factors	Price sensitivity, delivery doubts, mistaken orders	Pre-purchase confirmation and pricing strategy
Returns	Product quality & delivery speed	Specification mismatch, quality defects	Quality assurance and packaging standards

8.5 Monthly Cancellation Trend

Cancellations are highly fluctuating, with no consistent upward or downward trend. June is a recurring high-cancellation month. The lowest cancellations were recorded in May 2024 and September 2024 (4 each). Large month-over-month spikes suggest possible issues related to seasonality, promotions, delivery delays, or customer satisfaction.

9. CUSTOMER BEHAVIOUR & REVENUE LOSS

9.1 Revenue Lost Due to Cancellations and Returns

Rs. 5,19,674	41.09%	Rs. 7,45,088
Total Revenue Lost	Lost Revenue %	Retained Revenue

Around 41.09% of total revenue is lost due to cancellations and returns. This is quite high and suggests the business should focus on reducing these orders to improve profits and achieve more stable growth.

9.2 Revenue Loss by Cancellations (Product-wise)

Product	Revenue Lost (Cancellation)	Note
Chair	Rs. 48,660.98	Highest — largest share of cancellation revenue loss
Laptop	Rs. ~39,000	2nd
Tablet	Rs. ~37,000	3rd
Others	—	Distributed across remaining products

Chair contributed the highest cancelled order value (Rs. 48,660.98), followed by Laptop and Tablet. These products account for the largest share of potential revenue loss from cancellations and should be analysed further to identify the underlying causes.

9.3 Revenue Loss by Returns (Product-wise)

Product	Revenue Lost (Return)	Note
Tablet	Rs. 42,525.86	Highest — largest share of return revenue loss
Monitor	Rs. 40,524.60	2nd
Laptop	Rs. 39,654.30	3rd
Others	—	Distributed across remaining products

Tablet contributed the highest returned order value (Rs. 42,525.86), followed by Monitor (Rs. 40,524.60) and Laptop (Rs. 39,654.30). These products account for the largest share of potential revenue loss due to returns and may require further investigation into product quality, customer expectations, or delivery-related issues.

One interesting observation: Chair had the highest cancellation revenue loss (Rs. 48,660.98), while Tablet had the highest return revenue loss (Rs. 42,525.86). This suggests that customers tend to cancel Chairs before receiving them, whereas customers tend to return Tablets after receiving them.

9.4 Top 5 Customers by Spending

Customer ID	Total Spending	Rank
C38840	Rs. 5,723.23	1st
C57276	Rs. 3,456.40	2nd
C67260	Rs. 3,390.80	3rd
C13877	Rs. 3,384.90	4th
C18404	Rs. 3,370.20	5th

9.5 Coupon Code Contribution to Returns

Coupon Code	Returns	Note
No Coupon	76	Highest — most returns occur without a coupon discount
WINTER15	63	2nd
FREESHIP	61	3rd
SAVE10	47	Lowest

No coupon has the highest number of returns. Most returns happen because the customer did not find the product satisfactory — product mismatch, quality issue, or anything which did not meet the customer's expectations.

10. BUSINESS RECOMMENDATIONS

The business has generated Rs. 1.26 million in total revenue with a healthy Average Order Value of Rs. 1,053.97, indicating strong customer spending. However, the company's biggest challenge is the high number of cancellations and returns, which account for 41.09% of total revenue loss. Reducing these losses should be the top priority, as even a small improvement can significantly increase profitability.

CRITICAL PRIORITY

Focus: Quality Control & Returns

Actions: Implement comprehensive QA audit for electronics. Establish enhanced packaging standards. Develop quality guarantee protocols specifically for Tablets, Laptops, and Printers which show the highest return rates.

Outcome: Reduce return rate by 15–20%. Improve customer satisfaction.

CRITICAL PRIORITY

Focus: Cancellation Management

Actions: Implement order verification for high-value items. Provide genuine customer reviews, product warranties, detailed product descriptions, and secure delivery assurances. Give special attention to Chair cancellations as it contributes the highest cancellation revenue loss (Rs. 48,660.98).

Outcome: Reduce cancellation rate by 10–15%. Improve customer commitment.

HIGH PRIORITY

Focus: Seasonal Planning & Execution

Actions: Prepare inventory levels for May–June peaks. Plan staffing and fulfilment capacity expansion. Launch advance marketing campaigns ahead of peak season. Since May–June consistently represent peak demand periods, this advance planning can capture 15–20% additional revenue.

Outcome: Capture 15–20% additional peak season revenue.

HIGH PRIORITY

Focus: Marketing Channel Optimisation

Actions: Leverage Instagram as the primary referral channel — it is the strongest-performing source generating the highest revenue and order volume. Continue investing in this channel and replicate successful strategies across other marketing platforms.

Outcome: Maintain sales momentum. Improve marketing ROI.

MEDIUM PRIORITY

Focus: Coupon & Promotional Strategy

Actions: The FREESHIP coupon generated the highest revenue, suggesting that free shipping promotions are effective in driving sales. Cancellations are fairly evenly distributed across coupon codes, suggesting coupons are not the primary cause of cancellations and greater focus should be placed on product and operational improvements.

Outcome: Expand effective promotional strategies to improve profitability.

MEDIUM PRIORITY

Focus: Product Portfolio Optimisation

Actions: Investigate Chair cancellation causes specifically. Enhance electronics product descriptions to reduce expectation mismatch. Create quality standards for Tablet and electronic deliveries. Verify that products match customer expectations before shipping.

Outcome: Improve overall customer satisfaction. Reduce expectation mismatch.

11. FINANCIAL IMPACT ANALYSIS

Reducing cancellation and return rates offers substantial financial opportunity. The analysis evaluates multiple improvement scenarios and their corresponding revenue impact.

Scenario	Cancellation Rate	Return Rate	Revenue at Risk	Revenue Retained	Net Improvement
Current State	20.83%	20.58%	Rs. 5.19L (41.09%)	Rs. 7.45L	Baseline
15% Reduction Target	17.70%	17.49%	Rs. 3.54L (27.98%)	Rs. 9.11L	+23% improvement
25% Reduction Target	15.62%	15.44%	Rs. 2.47L (19.54%)	Rs. 10.18L	+38% improvement
Best Case Scenario	10.00%	10.00%	Rs. 1.26L (10.00%)	Rs. 11.39L	+54% improvement

A 15% reduction in combined cancellation and return rates would increase annual revenue retention by approximately Rs. 1.72 Lakh, representing a 23% improvement. A 25% reduction would recover approximately Rs. 2.79 Lakh in annual revenue — a 38% improvement. These financial gains translate directly to bottom-line profitability and demonstrate the critical importance of addressing these metrics.

12-Month Implementation Targets

Metric	Current	12-Month Target	Expected Impact
Annual Revenue	Rs. 12.6L	Rs. 14.3L+	13–22% improvement
Cancellation Rate	20.83%	15–17%	Reduce by ~3–5 percentage points
Return Rate	20.58%	15–17%	Reduce by ~3–5 percentage points
Customer Satisfaction	Unknown	8/10 minimum	Establish and improve baseline
Operational Efficiency	Unknown	90%+ target	Quality control achievement

12. FINAL CONCLUSION

Business Strengths

- The business is generating strong revenue of Rs. 1.26 million with a healthy Average Order Value of Rs. 1,053.97.
- The balanced product portfolio with no single category dominating provides resilience and multiple revenue streams.
- Predictable seasonal patterns (May–June peaks) enable proactive planning.
- Instagram and FREESHIP have proven effectiveness as top marketing channel and promotion respectively.
- Price optimisation is viable due to relatively inelastic customer demand.

Critical Challenge

The 41.09% combined cancellation and return rate represents the organisation's primary strategic challenge. This issue dwarfs all other considerations and must become the top organisational priority. Unlike other metrics showing business strength, this metric indicates systematic operational issues that require immediate intervention.

Key Insights

1. Returns are product-quality driven (electronics-focused) — requires enhanced QA and packaging standards.
2. Cancellations are behaviour-driven across multiple factors — requires customer confidence improvement mechanisms.
3. Chairs show the highest cancellations; customers tend to cancel before receiving them.
4. Tablets show the highest returns; customers tend to return them after receiving them.
5. Seasonal peaks are predictable — enables advance resource and campaign planning for May–June.
6. Successful marketing channels exist — Instagram and FREESHIP demonstrate proven effectiveness.
7. Price optimisation is viable — strategic pricing will not significantly reduce purchase volume.

Recommended Priority: Reducing cancellations and returns will yield the greatest impact on profitability and long-term sustainability. Overall, the business has strong revenue potential, but improving revenue retention by reducing cancellations and returns will have the greatest impact on profitability and long-term growth. Implementation of the strategic recommendations outlined in this report can recover Rs. 1.72 Lakh to Rs. 2.79 Lakh in annual revenue while simultaneously improving customer satisfaction and brand reputation.

— End of Report —